

Soil Health & Crop Recommendation Report

Report Date: November 07, 2025 at 19:28:15 UTC

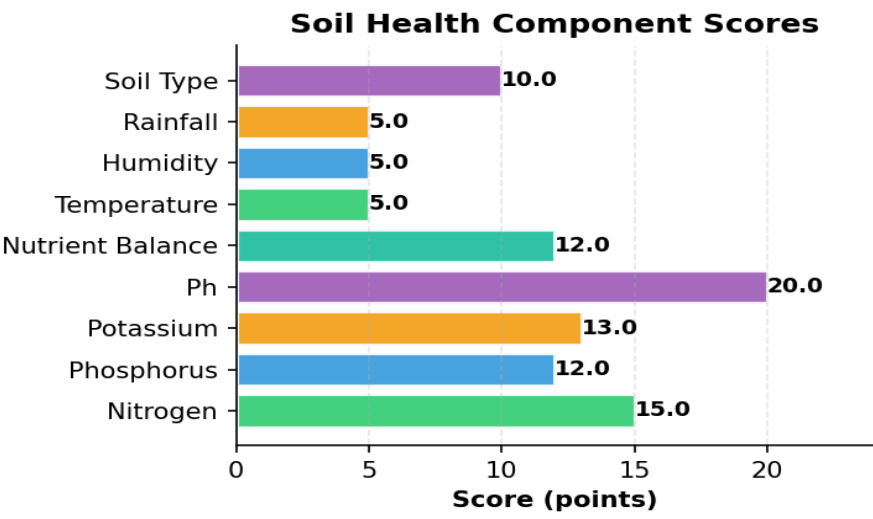
Farmer Information	
Name:	goli
Phone:	1234567891
Location:	vijayapura
Land Size:	1.00 acres

Parameter	Value	Unit
Nitrogen (N)	50.0	mg/kg
Phosphorus (P)	50.0	mg/kg
Potassium (K)	50.0	mg/kg
pH Level	6.50	
Temperature	25.0	°C
Humidity	60.0	%
Rainfall	120.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score	97.0/100	Excellent	
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Component	Score	Points
Nitrogen	15.0	points
Phosphorus	12.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	12.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	5.0	points
Soil Type	10.0	points



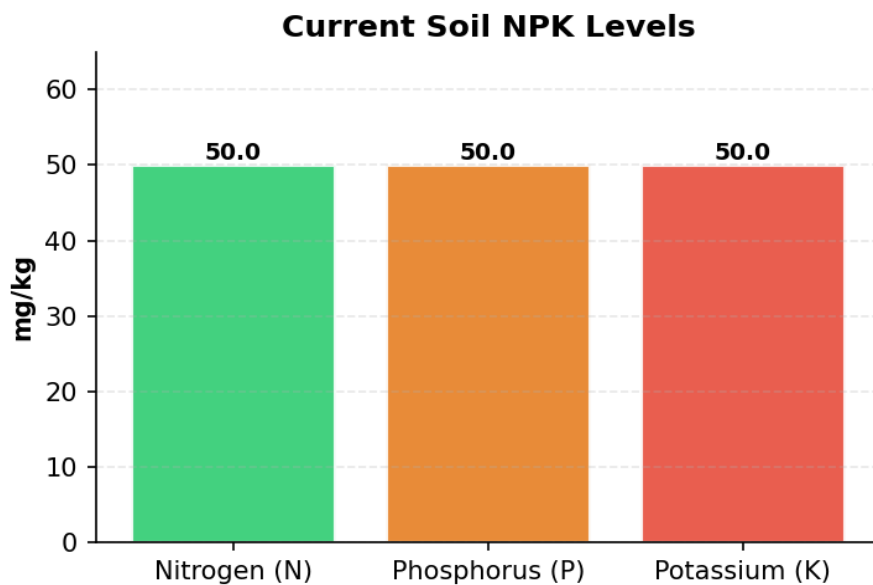
Recommended Crop

Predicted Crop	Jute
Confidence Level	23.9%

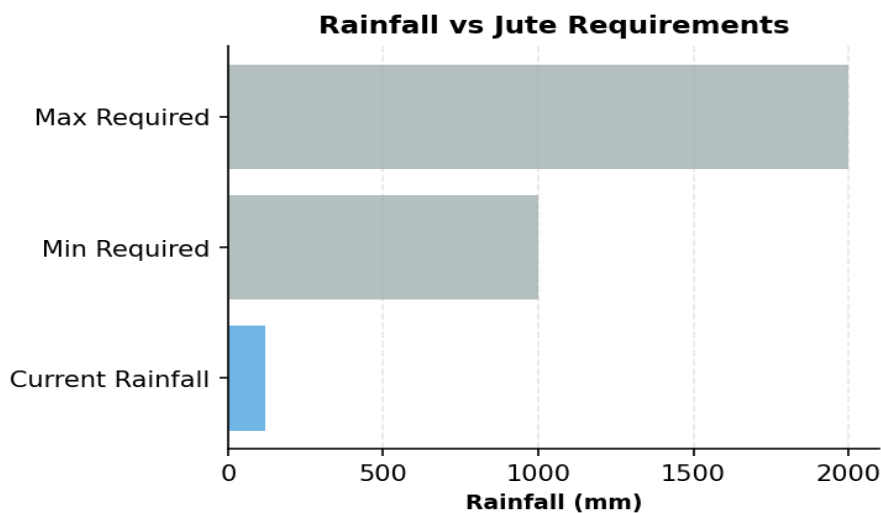
Alternative Crop Options

Rank	Crop	Confidence (%)
1	Pigeonpeas	19.7%
2	Banana	13.7%

Soil NPK Levels Visualization



Rainfall Analysis



Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for jute.
- Nitrogen (50.0 mg/kg) is within optimal range for jute.
- Phosphorus (50.0 mg/kg) is excessive for jute. Optimal range: 20-40. High P can interfere with micronutrient uptake.
- Potassium (50.0 mg/kg) is excessive for jute. Optimal range: 20-40. High K can cause magnesium deficiency.
- Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

Fertilizer Requirements for jute:

- Land Size: 1.0 acres
- Current NPK: N=50.0, P=50.0, K=50.0 mg/kg
- Ideal NPK: N=60, P=30, K=30 kg/acre

Required Application:

- Nitrogen (N): 10.0 kg total (10.0 kg/acre)
- Apply in 2-3 split doses: 50% at sowing, 25% at tillering, 25% at flowering

Organic Alternatives:

- Compost/FYM: 5-10 tons/acre (provides N, P, K)
- Green Manure: Leguminous crops (fixes N)
- Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Jute

Requires high humidity and consistent moisture. Harvest before flowering for best fiber.

Additional Notes

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.

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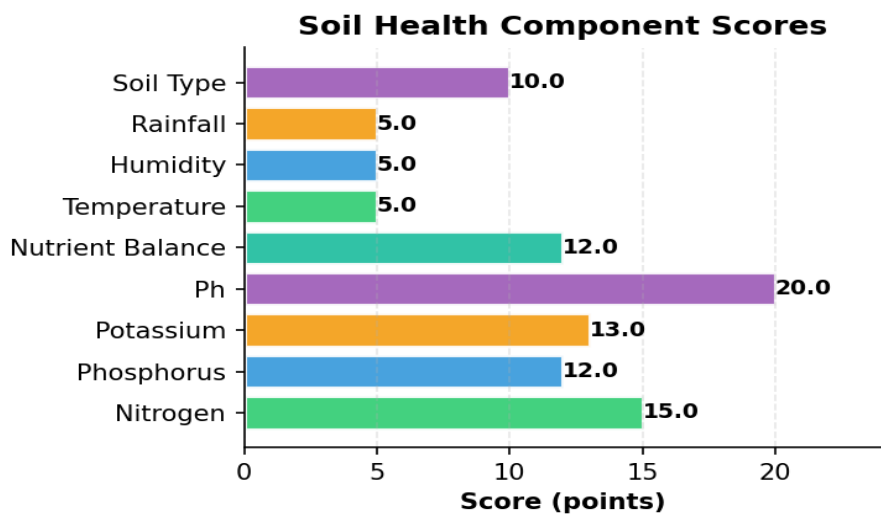
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Soil Health & Crop Recommendation System

Comprehensive Agricultural Analysis Report



Recommended Crop

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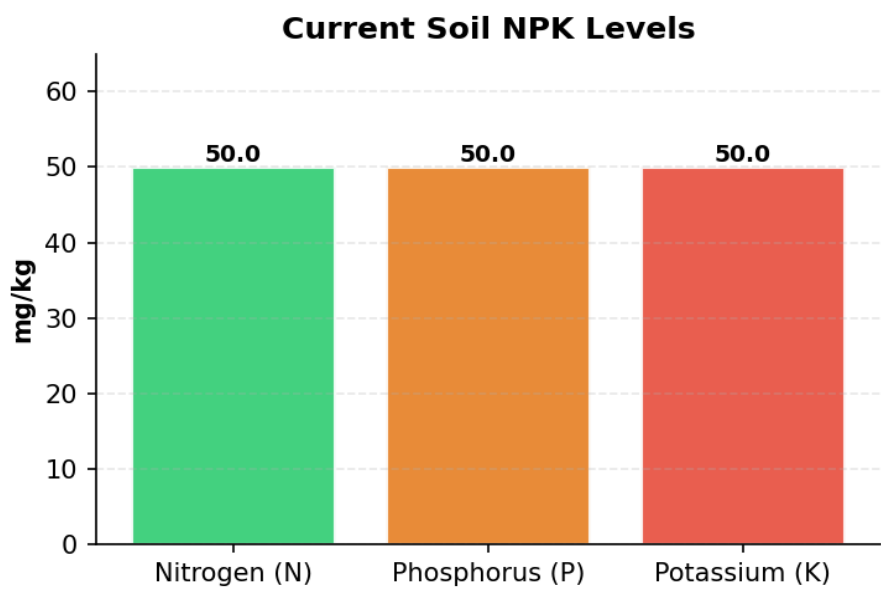
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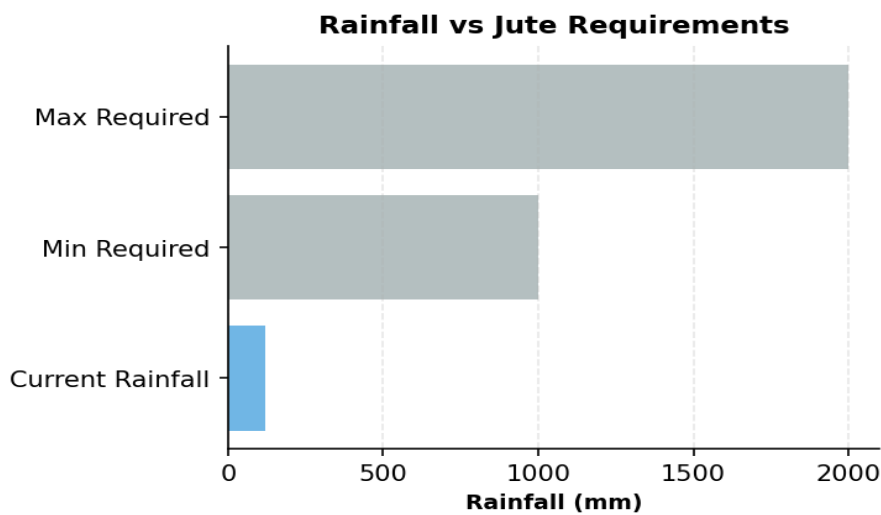
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